

Banka Venkateswarlu

Aspiring AI Engineer & Data Scientist

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EDUCATION

Bachelor of Technology in Computer Science and Engineering
Vel Tech University, Chennai

08/2022 – 06/2026
CGPA: 8.49

Diploma in Electronics and Communication Engineering
Government Polytechnic College, Vijayawada

07/2019 – 07/2022
Percentage: 82.18%

PROJECTS

1. Scientific Literature Intelligence System

RAG-Based Biomedical Research Assistant

- Built a biomedical RAG application using LangChain, FAISS, BioBERT embeddings, and Llama 3.1 for scientific literature intelligence.
- Implemented paper-isolated FAISS vector databases to eliminate cross-document context contamination and improve retrieval accuracy.
- Developed semantic scientific chunking and cross-paper reasoning pipelines for summarization, comparative analysis, and conversational QA.
- Integrated BioBERT-based sentence embeddings for domain-specific semantic search across biomedical research papers.
- Designed an interactive Streamlit UI supporting multi-PDF upload, semantic retrieval, structured comparison, and research-focused insights.

2. Loan Status Prediction and Customer Clustering

Machine Learning Classification & Clustering Project

- Built a loan approval prediction model using KNN classification with data preprocessing and feature scaling.
- Performed exploratory data analysis and handled missing values and outliers for reliable predictions.
- Applied K-Means clustering to segment customers based on financial and demographic attributes.
- Analyzed cluster patterns to identify customer groups and support data-driven loan decision insights.

3. PPE Kit Detection using YOLOv8

Deep Learning Computer Vision Project

- Developed an AI-powered Personal Protective Equipment (PPE) detection system using YOLOv8 for construction site safety.
- Trained the model to detect safety helmets, safety vests, gloves, boots, and workers in real time.
- Performed image annotation, preprocessing, model training, and evaluation using the Ultralytics YOLOv8 framework.
- Achieved high detection accuracy and enabled automated PPE compliance monitoring to improve workplace safety.

SKILLS

Programming Languages: Python, SQL

Machine Learning & Deep Learning: Machine Learning, CNN, RNN Classification, Clustering

Generative AI: RAG, LangChain Basics, Ollama

Libraries & Frameworks: Scikit-learn, TensorFlow/Keras, Pandas, NumPy, Streamlit